Assignment 1A: Multi-Armed Bandits

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Group 55

1 Introduction

Reinforcement Learning (RL) is a computational framework for automating goal-directed learning through direct interaction with an environment. At its core, RL involves learning how to map situations to actions to maximize a numerical reward signal. This assignment focuses on the "Multi-Armed Bandit" problem, a classic RL framework that encapsulates the "exploration-exploitation trade-off." This trade-off represents the fundamental conflict between choosing a known, high-reward action (exploitation) and trying new, potentially better options to gather more data (exploration). An agent in an unknown environment must balance these two needs: it must explore long enough to ensure its observed outcomes accurately reflect the true reward distributions, yet it must avoid spending excessive time on sub-optimal actions to ensure cumulative rewards are maximized. To address this, we evaluate and compare three distinct strategies: the e-Greedy agent, the Optimistic Initialization (OI) agent, and the Upper Confidence Bound (UCB) agent.

2 The Environment

The environment for Assignment 1A simulates a movie recommendation system modeled as a 25-armed bandit problem. The agent performs 1.000 steps per run, aiming to identify which of the 25 possible actions (representing movie recommendations) yields the highest average reward. Interaction occurs via the `step()` function. In each iteration, the agent selects an action according to its active policy, and the environment returns a simulated reward, representing a movie rating, sampled from a normal distribution specific to that action. Central to the agent's decision-making is the Q-value, $Q(a)$, which represents the current estimate of the expected reward for each action. These estimates are stored in a one-dimensional array of size 25. The Q-values are refined through update rules, which provide the computational mechanism for the agent to adjust its internal knowledge based on the rewards observed through experience.

3 The Three Policies

- **E-Greedy Agent:** The E-Greedy policy, first policy we encounter, is a simple yet effective way of managing the exploration-exploitation trade-off. The balance is governed by the parameter ϵ ($0 \leq \epsilon \leq 1$), as defined by the following policy:

$$\pi_{\epsilon-greedy}(a) = \begin{cases} 1-\epsilon, & \text{if a} = \text{argmax}_{b\mathcal{A}} Q(b) \\ \frac{\epsilon}{|\mathcal{A}|-1}, & \text{otherwise} \end{cases}$$

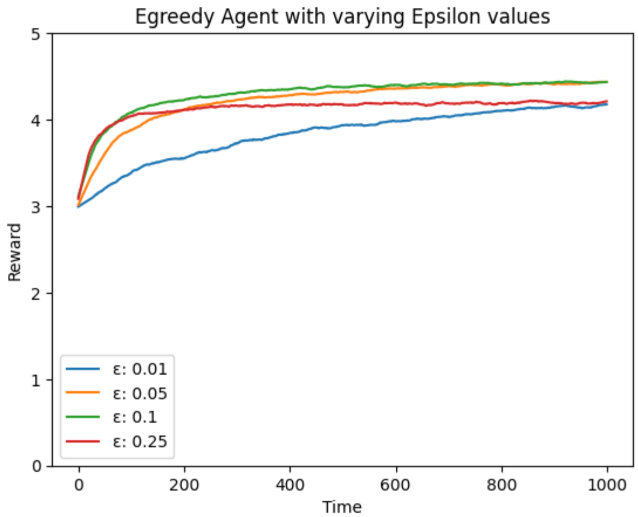
Under this mechanism, the agent exploits its current knowledge with probability $1 - epsilon$ by selecting the action with the highest estimated value. Conversely, with probability ϵ, it explores by choosing an action at random. This ensures that the agent continues to gather data on potentially superior actions rather than stagnating on a sub-optimal choice. A primary advantage of e-Greedy is its simplicity and the guarantee that, given sufficient time, the agent will sample every action. However, its exploration is entirely stochastic; because it selects actions randomly without considering their potential, it may waste steps on actions that have already proven to be poor. To refine these estimates over time, we transition from a simple overwrite method to an incremental update rule:

$$Q(a) \leftarrow Q(a) + \frac{1}{n(a)}[r(a) - Q(a)]$$

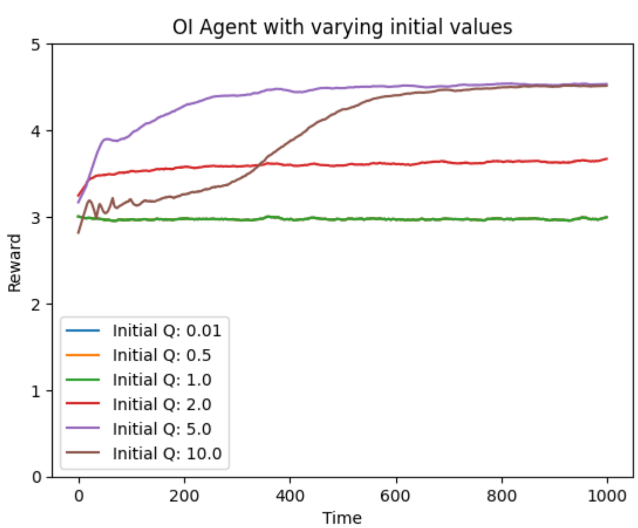
In this sample-average formulation, \$n(a)\$ represents the step count for action \$a\$. As \$n(a)\$ grows, the weight of new rewards decreases, allowing the estimate to converge to the true arithmetic mean. Alternatively, a constant step-size parameter α can be utilized:

$$Q(a) \leftarrow Q(a) + \alpha[r(a) - Q(a)]$$

The fundamental difference lies in temporal weighting. While the \$1/n(a)\$ rule weights all historical rewards equally, the α parameter gives higher priority to recent observations, causing the influence of older data to decay exponentially. This makes the α update rule particularly effective in non-stationary environments where reward distributions change over time.



- **Optimistic Initialization (OI) Agent:** Optimistic Initialization is an alternative exploration strategy that, unlike the e-Greedy approach, manipulates the starting state of the agent's knowledge to benefit exploration. By initializing Q-values to a significantly high constant (e.g., $Q(a) = 5$ or 10) rather than the standard zero, we fundamentally bias the agent's initial expectations for every available action. This strategy induces systematic exploration through a "disappointment" mechanism. When the agent selects an action greedily, it expects a high reward based on its optimistic initialization. However, because the actual reward provided by the environment is much lower than this estimate, the incremental update rule forces the Q-value for that action to decrease toward its realistic mean. Consequently, the previously selected action loses its status as the "highest valued," prompting the greedy algorithm to switch to a different, untried action that still retains its optimistic starting value.

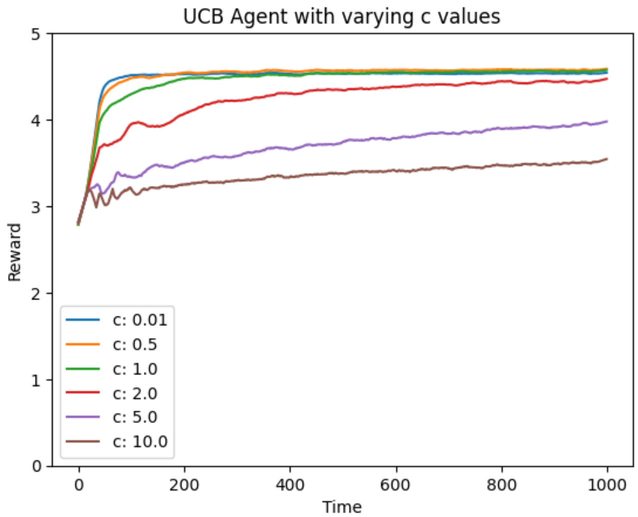


The performance plots highlight the impact of these initial settings. When using low initial values, the learning curve ascends slowly because the agent lacks an inherent incentive to deviate from its first successful discovery. In contrast, high initial values force the agent to perform an exhaustive initial sweep of all actions. This ensures the true optimal action is identified early, leading to a much sudden improvement in cumulative rewards as the estimates converge. This approach compels the agent to exhaustively explore all actions at the start of the simulation, maintaining a cycle of discovery until every option has been sampled multiple times. However, the efficacy of the strategy is highly sensitive to the chosen initial value. Setting the value too high leads to excessive, inefficient exploration, whereas a value that is too low fails to activate the "disappointment" mechanism required to drive the agent toward untried actions.

- **Upper Confidence Bounds (UCB) Agent:** UCB, the third policy, has a deterministic exploration strategy. Unlike the e-Greedy, that explores actions at random, UCB uses uncertainty to explore in a more targeted manner.
The policy can mathematically be formulated as:

$$\pi_{UCB}(a) = \begin{cases} 1, & \text{if a} = \text{argmax}_a[Q_t(a) + c \cdot \sqrt{\frac{ln(t)}{n(a)}}] \\ 0, & \text{otherwise} \end{cases}$$

where t is the timestep, c is the exploration constant and $n(a)$ the action counter. The term $n(a)$ is in the denominator, which means that as $n(a)$ decreases, the value of the fraction $\frac{ln(t)}{n(a)}$ increases (and vice versa). This brings the exploration term $c \cdot \sqrt{\frac{ln(t)}{n(b)}}$ to become larger. Therefore, if an action has been tried very few times, its $n(a)$ will be small, leading to a large exploration constant. This large constant increase the overall UCB value for that action, making it more likely to be selected even if its current estimated reward $Q(a)$ isn't the highest. A crucial aspect of UCB is, other than exploring actions that haven't been selected often, how it handles actions that have never been tried. If $n(a) = 0$ the exploration term would involve a division by 0. To address this. we treat the estimate of that action as infinite. This ensures that before start comparing the rewards, every available action is tried at least once.



4 Comparison

The following plot illustrates a performance comparison between the optimal hyperparameters identified for each of the three policies. This visualization tracks the evolution of the average reward across all timesteps. While all three strategies converge to a similar performance level by timestep 1.000, with the e-Greedy policy trailing slightly, the most significant deviations occur within the first 300 steps. The growth rates vary markedly during this initial phase. UCB demonstrates the most rapid ascent; by evaluating the potential of each action through a combination of estimated value and uncertainty, it identifies high-reward actions more efficiently than its counterparts. In contrast, the trajectories for Optimistic Initialization (OI) and e-Greedy are more gradual.

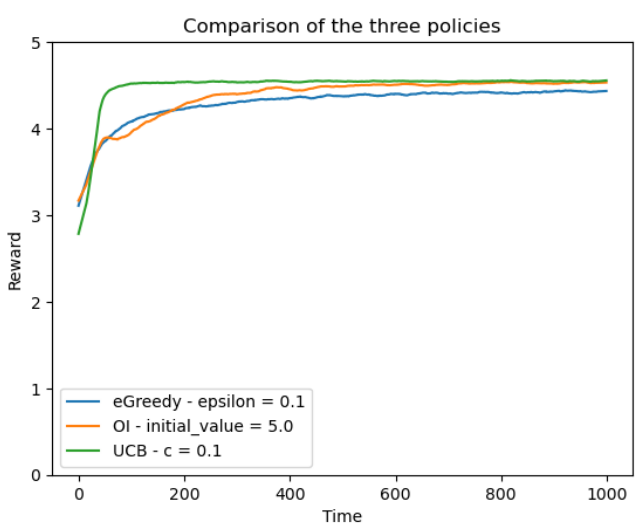


Figure 1: Comparison plot 1

The OI policy exhibits more early-stage oscillation and initially slightly underperforms relative to e-Greedy. However, as the "disappointment" mechanism of OI systematically eliminates sub-optimal actions, its performance surpasses the e-Greedy curve around timestep 200, eventually yielding a higher average reward.
The Second Comparison Plot, instead, focuses on displaying the average overall performance of each hyperparameter for each of the three policies. This visualization reveals that the peak performance for all three strategies is concentrated within a narrow range between 4.2 and 4.5. These results suggest that no single policy is inherently superior in this environment. Instead, the data underscores that achieving high-fidelity performance depends less on the choice of algorithm and more on the precise calibration of exploration parameters. Finding the "sweet spot" for ϵ, c, or the initial Q-value is the primary factor in maximizing the agent's efficiency.

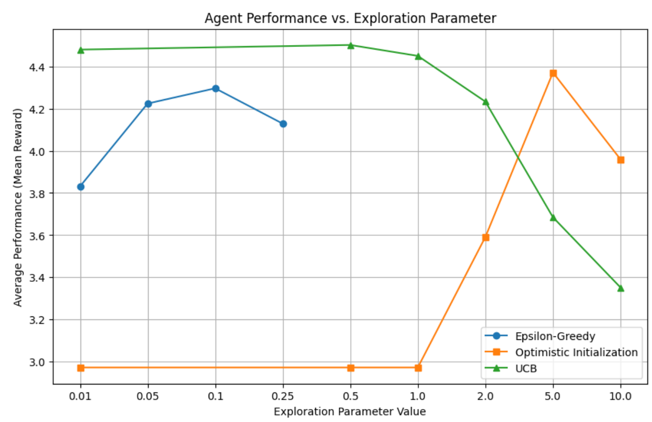


Figure 2: Comparison Plot 2

5 Evaluation

Experimental results indicate that while e-Greedy, Optimistic Initialization, and UCB strategies achieve similar convergence, UCB offers superior optimization speed by systematically accounting for uncertainty. This efficiency makes UCB the preferred choice for environments requiring the rapid identification of optimal actions during the early stages of interaction. A critical takeaway is that performance often depends more on the precise calibration of exploration hyperparameters than on the specific algorithm selected. Therefore, to maximize cumulative rewards, you should prioritize finding the best value for these parameters to prevent the agent from over-exploring sub-optimal options.